HTTPS flow

MCC1

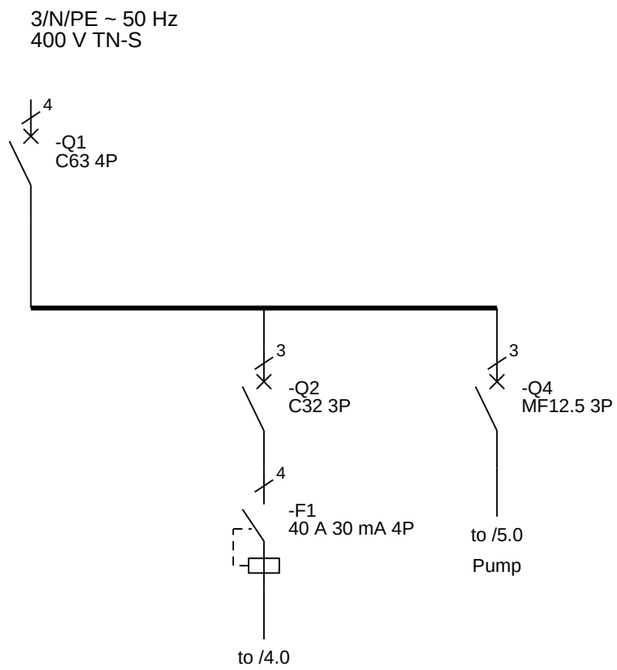
Company profile	iec-default
Design rules	software defaults, not confirmed by the company's engineers

Rev.	Date	Description	Drawn	Checked	Review approved
01	2026-10-31				
02	2026-10-31				Synthetic Reviewer

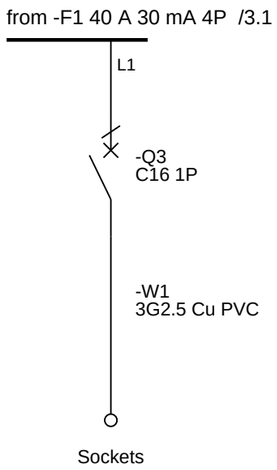
Company	Project HTTPS flow	Job number	Board	Title of page **Title page**	Rev. 02
Date 2026-10-31	Drawn	Checked	Review approved Synthetic Reviewer	Page 1 of 11	

Page	Title	Board
1	Title page	
3	Main power	MCC1
4	Distribution loads	MCC1
5	Distribution loads	MCC1
6	Design notes	MCC1
7	Layout	MCC1
8	Terminals	MCC1
9	Cable list	
10	Parts list	

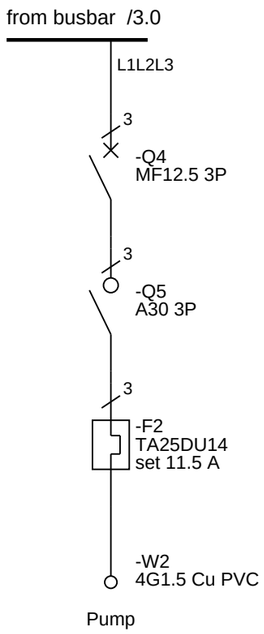
Company	Project	Job number	Board	Title of page	Rev.
	HTTPS flow			**Table of contents**	02
Date	Drawn	Checked	Review approved	Page	
2026-10-31			Synthetic Reviewer	2 of 11	



Company	Project	Job number	Board	Title of page	Rev.
	HTTPS flow		MCC1	**Main power**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-31			Synthetic Reviewer	3 of 11	



Company	Project	Job number	Board	Title of page	Rev.
	HTTPS flow		MCC1	**Distribution loads**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-31			Synthetic Reviewer	4 of 11	

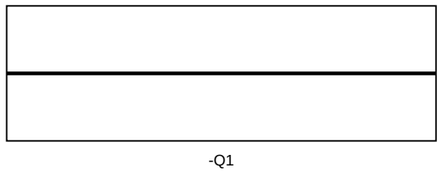


Company	Project	Job number	Board	Title of page	Rev.
	HTTPS flow		MCC1	**Distribution loads**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-31			Synthetic Reviewer	5 of 11	

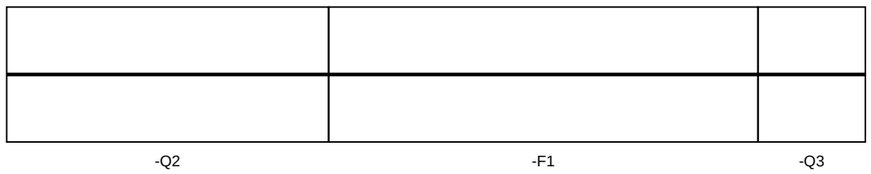
No.	Note
1	Group 1 breaker raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination.
2	Pump: Ir 11.5 A, the typical Ir of a 5.5 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate.
3	Pump: dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2.
4	Phase imbalance 55.6 % exceeds the company's 10 %: the loads cannot be spread more evenly as given.
5	Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.
6	No fault level given: every breaker's breaking capacity is left to be confirmed.
7	Leave 1 spare outgoing ways (20 %).
8	Loads without a power factor were taken at cos phi 0.9.
9	Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.
10	2 circuit(s) have no cable length given; their voltage drop is not checked.
11	No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze.
12	Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.
13	Circuit rules are this software's defaults, not confirmed by the company's engineers.
14	Enclosure: none named, since 3 devices have no width (rail_widths_mm in the company settings).

Company	Project	Job number	Board	Title of page	Rev.
	HTTPS flow		MCC1	**Design notes**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-31			Synthetic Reviewer	6 of 11	

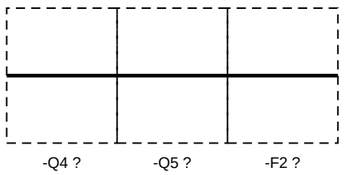
Incomer: 70 mm of rail



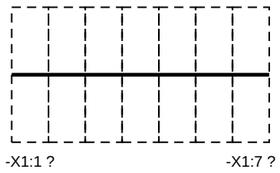
Residual current group: 140 mm of rail



Busbar: 0 mm of rail; 3 without a width



Terminals: 0 mm of rail; 7 without a width



Dashed: no width is given for the device's kind (rail_widths_mm in the company settings).

Company	Project	Job number	Board	Title of page	Rev.
	HTTPS flow		MCC1	**Layout**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-31			Synthetic Reviewer	7 of 11	

Terminal	Conductor	Circuit	Cable	mm²	Article
-X1:1	L1	Sockets	-W1	2.5	8WH1000-0AF00
-X1:2	N	Sockets	-W1	2.5	8WH1000-0AF00
-X1:3	PE	Sockets	-W1	2.5	8WH1000-0CF07
-X1:4	U	Pump	-W2	1.5	8WH1000-0AF00
-X1:5	V	Pump	-W2	1.5	8WH1000-0AF00
-X1:6	W	Pump	-W2	1.5	8WH1000-0AF00
-X1:7	PE	Pump	-W2	1.5	8WH1000-0CF07

Company	Project	Job number	Board	Title of page	Rev.
	HTTPS flow		MCC1	**Terminals**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-31			Synthetic Reviewer	8 of 11	

Cable	From	To	Type	Length
MCC1 -W1	MCC1 -Q3	Sockets	3G2.5 Cu PVC	
MCC1 -W2	MCC1 -Q4	Pump	4G1.5 Cu PVC	

Company	Project	Job number	Board	Title of page	Rev.
	HTTPS flow			**Cable list**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-31			Synthetic Reviewer	9 of 11	

Qty	Description	Article	Order number	Devices
1	Circuit-breaker C63 4P	not selected		-Q1
1	Circuit-breaker C32 3P	not selected		-Q2
1	Residual current circuit-breaker 40 A 30 mA 4P	not selected		-F1
1	Circuit-breaker MF12.5 3P	ABB T2S160 MF 12.5		-Q4
1	Contactor A30 3P	ABB A30		-Q5
1	Overload relay TA25DU14 set 11.5 A	ABB TA25DU14		-F2
1	Circuit-breaker C16 1P	not selected		-Q3
5	Terminal 8WH1	Siemens 8WH1000-0AF00	8WH1000-0AF00	MCC1 -X1
2	PE terminal 8WH1	Siemens 8WH1000-0CF07	8WH1000-0CF07	MCC1 -X1

Company	Project	Job number	Board	Title of page	Rev.
	HTTPS flow			**Parts list**	02
Date	**Drawn**	**Checked**	**Review approved**	**Page**	
2026-10-31			Synthetic Reviewer	10 of 11	

Findings (0 blocking, 8 warning, 6 info), page 1 of 1 — sheet 11 of 11

[WARNING] MCC1 / c1: Pump: Ir 11.5 A, the typical Ir of a 5.5 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source: policy; engineering confirmation required)
[WARNING] MCC1: Phase imbalance 55.6 % exceeds the company's 10 %: the loads cannot be spread more evenly as given. (phase_imbalance; source: company max_phase_imbalance_percent; engineering confirmation required)
[WARNING] MCC1: No fault level given: every breaker's breaking capacity is left to be confirmed. (no_fault_level; source: policy; engineering confirmation required)
[WARNING] MCC1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)
[WARNING] MCC1: 2 circuit(s) have no cable length given; their voltage drop is not checked. (voltage_drop_unchecked; source: policy; engineering confirmation required)
[WARNING] MCC1: No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze. (earth_fault_no_ze; source: policy; engineering confirmation required)
[WARNING] MCC1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)
[WARNING] MCC1: Enclosure: none named, since 3 devices have no width (rail_widths_mm in the company settings). (enclosure_unknown_widths; source: policy; engineering confirmation required)
[INFO] MCC1: Group 1 breaker raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)
[INFO] MCC1 / c1: Pump: dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2. (starter_table; source: policy)
[INFO] MCC1: Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination. (discrimination_incomer_raised; source: policy)
[INFO] MCC1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)
[INFO] MCC1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)
[INFO] MCC1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)